

CentralDogma- Δ CC: Disentangling Shared and Modality-Specific Variant Mechanisms across Genomic and Protein Language Models

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Abstract

Genomic language models (e.g. Evo 2) and protein language models (e.g. ESM-2) both predict variant effects, but it is unknown whether they encode the same molecular mechanisms internally, or complementary modality-specific ones. We introduce CentralDogma- Δ CC, a sparse cross-architecture, cross-modality crosscoder that operates not on raw activations but on paired variant-induced activation deltas: for each missense variant we compute $\Delta h_{\text{DNA}} = h_{\text{DNA}}(\text{mut}) - h_{\text{DNA}}(\text{wt})$ in Evo 2 and Δh_{PROT} in ESM-2, and learn shared and modality-private sparse latents that reconstruct both. On the BRCA1 saturation-genome-editing benchmark (2086 missense variants with deep mutational scanning function scores) the shared code (i) aligns the two models well enough to retrieve the paired variant across modalities far above chance (Recall@1 – vs. CCA –), (ii) predicts DMS function scores under a residue-disjoint split competitively with a concatenation baseline while using 64 interpretable dimensions, and (iii) yields latents that are enriched for protein domains and loss-of-function. Causally ablating a shared latent’s decoder direction inside both models moves their variant scores together, evidence that the latent captures a genuinely shared mechanism rather than a correlational artifact. We release code, activations, and the full evaluation harness.

1 Introduction

Foundation models now span the central dogma: Evo 2 [Brixi et al., 2026] models DNA at single-nucleotide resolution, while ESM-2 [Lin et al., 2023] models proteins. Both assign scores to human coding variants, and both are used for variant effect prediction (VEP). Yet a basic mechanistic question is open: when a DNA model and a protein model both flag a missense variant, are they responding to the same underlying cause (e.g. disruption of a conserved structural residue), or to different, modality-specific causes (splicing and codon context on the DNA side; folding and solvent exposure on the protein side)? Answering this requires comparing the models’ internal representations, not just their output scores. This is hard because the two models differ in architecture (StripedHyena vs. Transformer), tokenization (nucleotides vs. amino acids), input length, hidden width, and training objective; naively aligning raw activations is dominated by global nuisance factors such as gene identity and conservation.

We make three observations. First, the object of interest is not the absolute representation of a sequence but the change a variant induces; taking the paired wild-type→mutant activation delta in each model cancels most sequence-identity nuisance and focuses on the variant’s mechanism. Second, sparse dictionary methods (SAEs, crosscoders) provide an interpretable, causally manipulable basis; crosscoders in particular were designed to diff representations across models [Lindsey et al., 2024]. Third, restricting to a single gene (BRCA1) removes the dominant gene-identity shortcut, so any cross-modal alignment the model finds must reflect within-gene variant structure.

Combining these, CentralDogma- Δ CC is a crosscoder over paired deltas with an explicit shared / DNA-private / protein-private factorization. Our contributions:

- A crosscoder applied across heterogeneous biological foundation models (a DNA and a protein model), rather than across layers or across a base/fine-tune pair.
- Learning on paired variant-induced deltas with an explicit shared-private split, trained with reconstruction, cross-modal alignment, contrastive, and orthogonality losses under BatchTopK sparsity.

- A rigorous BRCA1 evaluation—cross-modal retrieval, residue- and domain-disjoint DMS and ClinVar prediction, annotation enrichment, and a cross-model causal ablation that perturbs the same latent inside both Evo 2 and ESM-2.

2 Related Work

Genomic and protein language models. Evo 2 [Brix et al., 2026] is a StripedHyena genomic model with strong zero-shot VEP, including on BRCA1 SGE data; ESM-2 [Lin et al., 2023] is a protein masked-LM whose representations encode structure and function. Sparse features and crosscoders. Sparse autoencoders recover interpretable features in LMs and, via InterPLM [Simon and Zou, 2025], in protein LMs. Crosscoders [Lindsey et al., 2024] extend SAEs to jointly encode multiple representation spaces and were introduced for cross-layer and model-diffing analysis; recent work applies them to cross-architecture diffing [Jiralerspong and Bricken, 2026] and to narrow fine-tuning [Kassem et al., 2026, Minder et al., 2025]. We instead diff across the central dogma—a DNA and a protein model of the same variant. Variant effect prediction. DMS assays such as BRCA1 SGE [Findlay et al., 2018] provide continuous functional labels; we use them not as a leaderboard target but to test whether shared latents carry mechanism.

3 Method

Paired variant deltas. For a missense SNV we build an 8,192 bp variant-centred window and score wild-type and mutant sequences with Evo 2, extracting the residual-stream activation at layer ℓ_{DNA} and SNV position; the DNA delta is $\Delta h_{\text{DNA}} = h^{\text{mut}} - h^{\text{wt}} \in \mathbb{R}^{d_{\text{DNA}}}$ ($d_{\text{DNA}}=4096$). Independently we substitute the residue in the protein and take the ESM-2 hidden-state delta $\Delta h_{\text{PROT}} \in \mathbb{R}^{d_{\text{PROT}}}$ ($d_{\text{PROT}}=1280$) at the variant residue and layer ℓ_{PROT} . Deltas are standardized per dimension on the training split.

Shared-private crosscoder. Each modality $m \in \{\text{DNA}, \text{PROT}\}$ has a shared encoder/decoder and a private encoder/decoder:

$$z_s^m = \text{ReLU}(E_s^m(\Delta h_m - b_m)), \quad z_p^m = \text{ReLU}(E_p^m(\Delta h_m - b_m)), \quad \widehat{\Delta h}_m = D_s^m \tilde{z}_s^m + D_p^m \tilde{z}_p^m + b_m,$$

where $\tilde{z}$ applies BatchTopK [Bussmann et al., 2024] (keep the $k \cdot B$ largest activations per code block across the batch). Shared codes of the same variant are aligned across modalities. The training objective is

$$\mathcal{L} = \underbrace{\text{NMSE}_{\text{DNA}} + \text{NMSE}_{\text{PROT}}}_{\text{reconstruction}} + \lambda_a \|\tilde{z}_s^{\text{DNA}} - \tilde{z}_s^{\text{PROT}}\|^2 + \lambda_c \mathcal{L}_{\text{InfoNCE}}(z_s^{\text{DNA}}, z_s^{\text{PROT}}) + \lambda_o (\text{Cov}(z_s^{\text{DNA}}, z_p^{\text{DNA}}) + \text{Cov}(z_s^{\text{PROT}}, z_p^{\text{PROT}})),$$

with $\tilde{z}$ the ℓ_2 -normalized shared code. The InfoNCE term makes row i of the DNA shared code retrieve row i of the protein shared code; because all variants are in BRCA1, negatives are in-gene and retrieval cannot exploit gene identity. Orthogonality decorrelates shared from private latents so that private codes capture modality-specific structure.

Causal intervention. For shared latent f , its DNA decoder column $D_s^{\text{DNA}}[:, f]$ (rescaled to raw activation units) is a direction in Evo 2’s residual stream; the protein column is a direction in ESM-2’s. During a variant forward pass we subtract the latent’s reconstructed contribution at the variant position and re-score, in each model, comparing against a matched-norm random direction.

4 Experimental setup

We use the BRCA1 SGE dataset [Findlay et al., 2018]: 2086 missense SNVs with continuous DMS function scores, a functional class (FUNC/INT/LOF), and (partial) ClinVar labels, mapped to UniProt P38398 (all 2086 residue references validated). Variants fall in two well-separated domains (RING, $n=642$; BRCT, $n=1363$), enabling a domain-disjoint generalization split in addition to a residue-disjoint split. Evo 2 is evo2_7b and ESM-2 is esm2_t33_650M; we select layers $\ell_{\text{DNA}}=\text{blocks.24.mlp.l3}$ and $\ell_{\text{PROT}}=\text{layer 30}$ by

a DMS probe (§5). The crosscoder uses shared/private width 64/64 with BatchTopK $k=16$. Baselines: single-model Δ linear probe, concatenation, linear CCA, and PLS. All numbers are on held-out splits.

5 Results

Results will be inserted here.

6 Discussion and limitations

Our study is deliberately scoped to one gene to eliminate gene-identity confounds; whether shared latents transfer across genes and assays is the key next step (ProteinGym extension). The paired-delta design assumes a well-defined variant position in both models, which holds for missense SNVs but not for splice/indel variants, which we can only analyze through the DNA branch. Causal effects are measured on the models’ own likelihood scores, not on downstream phenotype.

7 Conclusion

Paired variant deltas plus a shared-private crosscoder expose sparse, causally active mechanisms shared between a genomic and a protein foundation model, and separate them from modality-specific ones—a step toward mechanistic, cross-modal interpretability of biological foundation models.

References

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